

DIAL ALERT Bias and Fairness Analysis

Capstone Project Step 5

Ethical review of model explanations subgroup performance uncertainty and mitigation tradeoffs

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Academic clinical decision support prototype. This audit does not establish clinical benefit, causal fairness, regulatory compliance, or readiness for deployment.

Executive Summary

The locked Step 4 random forest was audited on 21,354 sessions from 170 patients who were not used for model fitting, hyperparameter selection, calibration choice, or threshold selection. The model achieved test average precision 0.395, ROC AUC 0.852, and Brier score 0.062. The fairness analysis preserves the validation-selected threshold of 0.143 and resamples entire patients when estimating uncertainty.

Point estimates show different fairness concerns across available groups. The recorded-sex disparate impact ratio was 0.705 and the equalized odds gap was 0.076. Age groups had a disparate impact ratio of 0.541, while diabetes status had near-equal alert selection rates but a 11.0% sensitivity gap. Patient-bootstrap intervals were wide, so the audit identifies monitoring priorities rather than proving discrimination or fairness.

Training-sample reweighting improved the recorded-sex disparate impact ratio from 0.705 to 0.764 and reduced the equalized odds gap from 0.076 to 0.058, with average precision changing from 0.395 to 0.394. A separate sex-specific threshold experiment performed worse on the test set and is rejected. The recommended next step is external and prospective validation with purposeful collection of race, ethnicity, socioeconomic, gender-identity, symptom, intervention, and workflow data.

Rubric Evidence

Requirement	Evidence
Explainability	Global permutation evidence, aggregate Monte Carlo SHAP, synthetic SHAP and LIME local explanations, and synthetic-profile dependence plots are compared.
Limitations	Imbalance, patient and temporal leakage, overfitting, calibration, repeated sessions, external validity, and causal limits are addressed.
Bias audit	Recorded sex, age, diabetes, sex-by-age intersections, and patient-equal-weight sensitivity analyses are reported with patient-bootstrap intervals.
Fairness metrics	Demographic parity difference, disparate impact ratio, equal opportunity, false-positive-rate, equalized odds, and predictive parity gaps are calculated.
Mitigation	Outcome-by-sex reweighting and validation-tuned sex-specific thresholds are tested, including accuracy-fairness tradeoffs and governance risks.
Ethical AI	False-negative harm, alert fatigue, automation bias, privacy, accountability, drift, human oversight, and deployment gates are documented.

1 Audit Scope and Data Sufficiency

The fairness question is whether DIAL-ALERT distributes alerts and errors acceptably across clinically and socially relevant groups. It is not enough to ask whether overall discrimination is good. The audit therefore examines who receives alerts, who is missed, who receives false alerts, how reliable the probabilities are, and whether mitigation transfers from validation patients to new patients.

Attribute	Availability	Audit use	Important limitation
Recorded sex	Available	Primary protected-attribute audit	Binary recorded field does not represent gender identity or variation beyond F and M
Age	Available	Four prespecified age bands	Age groups simplify a continuous and time-varying characteristic
Race or ethnicity	Unavailable	Not auditable	Fairness across racial and ethnic groups cannot be assessed
Socioeconomic status	Unavailable	Not auditable	Diabetes and dialysis vintage are not treated as socioeconomic proxies
Diabetes status	Available	Clinical subgroup audit	Clinical risk factor, not a substitute for socioeconomic status

The source field is named gender but contains only F and M. This report describes it as recorded sex because it cannot represent gender identity. Race, ethnicity, and socioeconomic status are absent. Their fairness cannot be inferred from diabetes, dialysis vintage, geography, or another convenient proxy. Missing protected-attribute data are a substantive limitation and a requirement for future data collection.

Audit Population

Element	Specification
Test observations	21,354 sessions
Independent test units	170 patients
Patient overlap	Zero across training validation and test partitions
Operating rule	Global probability threshold 0.143 selected on validation patients
Uncertainty	500 bootstrap resamples of whole patients

2 Ethical Framing and Harm Model

DIAL-ALERT predicts a later systolic blood pressure below 90 mmHg. The target is an audible blood-pressure event, not a complete clinical diagnosis of intradialytic hypotension because symptoms, treatments, and clinician assessments are unavailable. A risk score should therefore support attention and preparation, not autonomously diagnose, prescribe fluid, change ultrafiltration, or terminate dialysis.

Concern	Meaning	Potential harm	Audit measure
False negative	An at-risk session is not flagged	Delayed recognition or preparation	Sensitivity and missed-event review
False positive	A low-risk session is flagged	Alarm fatigue unnecessary review or intervention	Precision false-positive rate and alert capacity
Unequal false negatives	One group is missed more often	Unequal safety benefit	Equal opportunity difference
Unequal false positives	One group receives more false alerts	Unequal burden and possible overtreatment	False-positive-rate difference
Unequal selection	Alert exposure differs regardless of outcome	Different access to review or intervention	Demographic parity and disparate impact
Miscalibration	The same score means different risk across groups	Misleading urgency and threshold decisions	Brier score and calibration error

No single fairness metric is sufficient. Different outcome prevalence and an imperfect classifier make some fairness goals mutually difficult to satisfy. Metric choice must follow a documented clinical harm model, workflow capacity, and community governance rather than a numerical pass-fail rule alone.

3 Explanation Framework

Four complementary explanation methods were used because each answers a different question. Agreement across methods strengthens confidence in a descriptive model interpretation, while disagreement or low local fidelity is itself evidence that an explanation should not be trusted quantitatively.

Method	Question answered	Main caution
Permutation importance	What information uniquely improves test-set ranking	Can understate correlated or redundant predictors
Approximate SHAP	How features distribute a prediction relative to background sessions	Monte Carlo and background dependent; correlated features complicate attribution
LIME-style surrogate	Which synthetic scenario features influence one nearby linear approximation	Only trustworthy when local fidelity is adequate
PDP	Average prediction change when one feature is varied	Can create unrealistic combinations and hides heterogeneity
Synthetic-profile ICE	How the same feature change affects predefined synthetic profiles	Illustrates model behavior, not observed patients or treatment effects

The SHAP implementation uses random background sessions and feature-order permutations to estimate interventional Shapley contributions. Only aggregate global importance is published. The local path ends at a manually constructed synthetic scenario, so no local explanation represents an actual HEMOBP session. Feature substitution can still form clinically implausible combinations. These values explain the fitted model, not physiology or causation.

4 Global Model Explanation

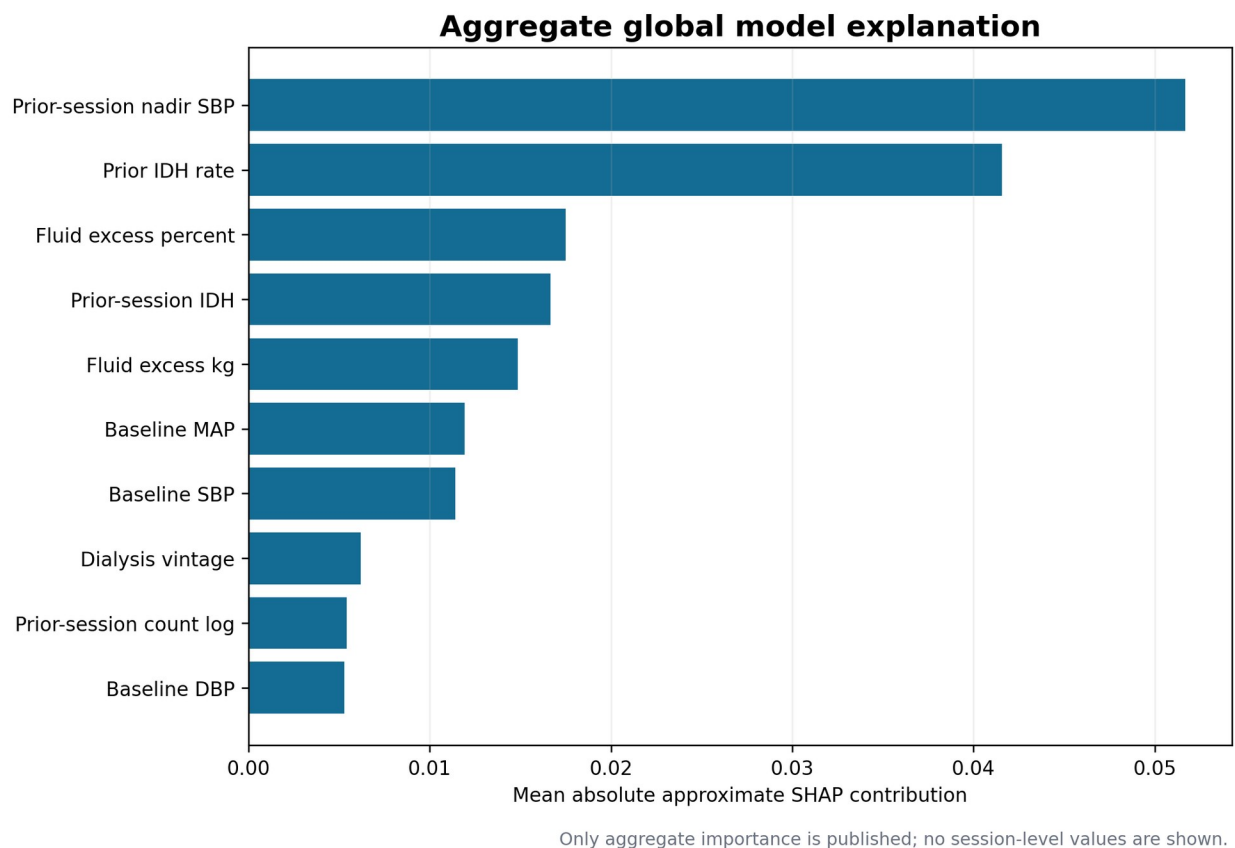


Figure 1. Aggregate mean absolute approximate SHAP contributions from an outcome-balanced analysis sample.

Feature	Mean absolute contribution	Mean signed contribution
Prior Nadir Sbp	0.052	+0.026
Prior Idh Rate	0.042	+0.025
Fluid Excess Pct	0.018	+0.009
Prior Session Idh	0.017	+0.007
Fluid Excess Kg	0.015	+0.009
Baseline Map	0.012	+0.006
Baseline Sbp	0.011	+0.006

Lower prior-session nadir SBP and higher prior IDH rate dominate risk predictions, followed by fluid excess and prior-session IDH. The same variables lead the independent permutation analysis. Recorded sex and age contribute little unique global information relative to the longitudinal haemodynamic features, but low feature importance does not rule out subgroup disparity through correlated predictors, threshold effects, or different baseline risk.

5 Synthetic Local Explanations

A predefined synthetic teaching scenario had a predicted probability of 0.691, compared with a background expectation of 0.083. The approximate SHAP contributions sum to the model prediction with local accuracy error $2.22\text{e-}16$. This scenario is explicitly marked `synthetic` and is not copied from a patient record. Prior IDH rate, prior-session nadir SBP, fluid excess, and a prior-session event produced the largest upward contributions.

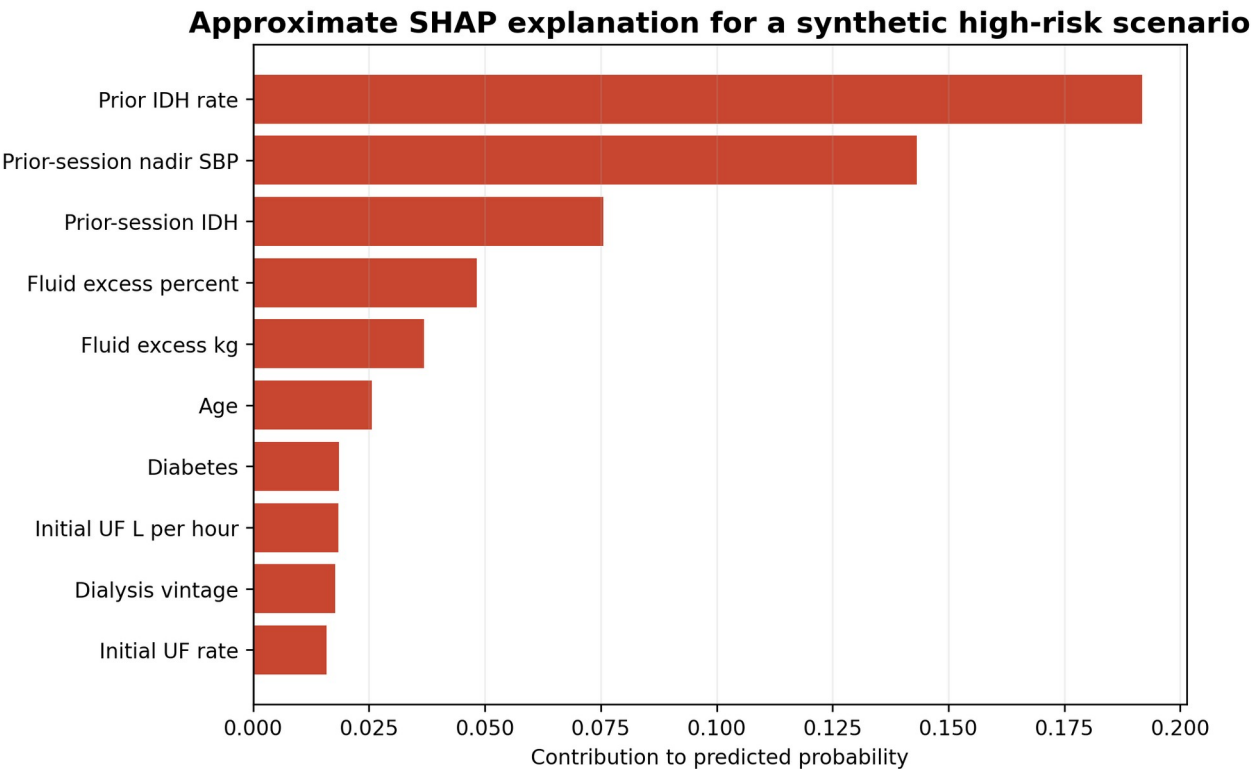


Figure 2. Approximate SHAP contributions for a synthetic high-risk teaching scenario.

The explanation is clinically plausible but not prescriptive. For example, the positive contribution from fluid excess does not prove that changing fluid removal would reduce risk. An actionable recommendation requires causal and prospective evidence that is outside this retrospective prediction study.

6 LIME Style Local Surrogate

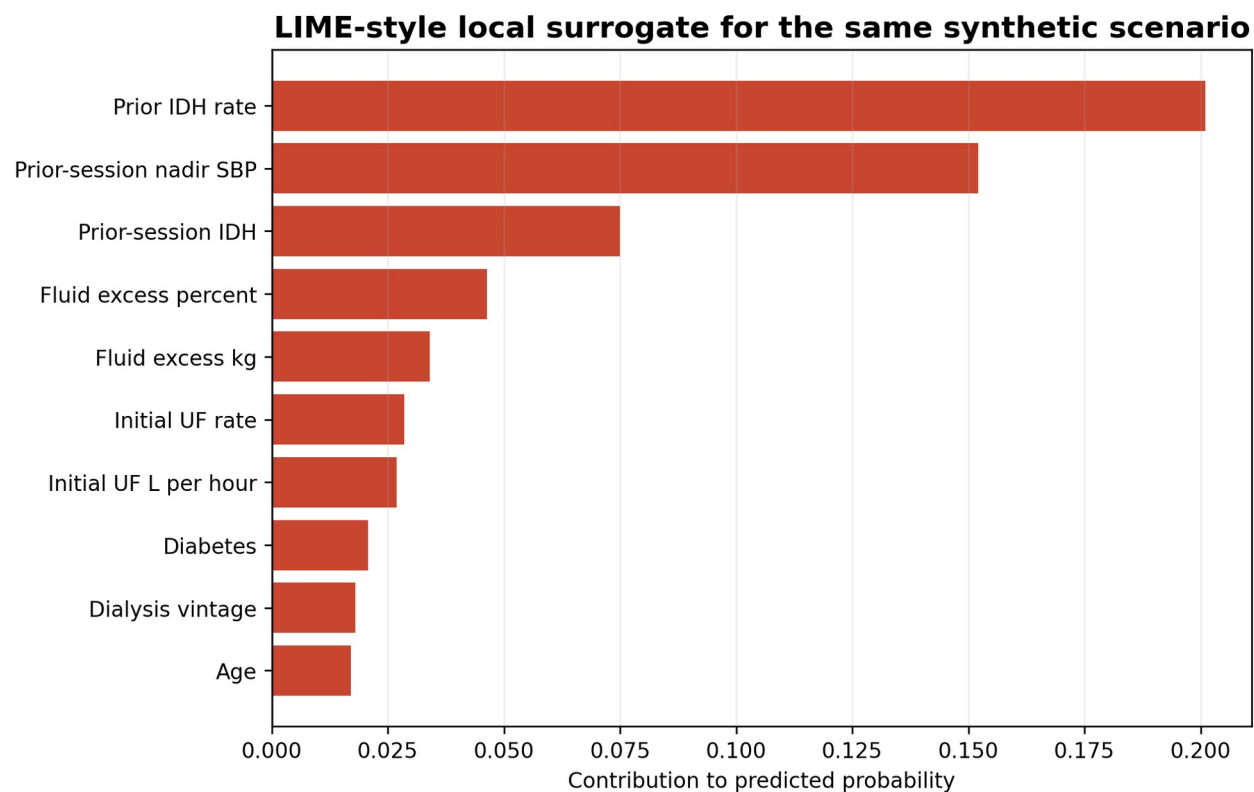


Figure 3. Weighted LIME-style linear surrogate for the same synthetic scenario.

Diagnostic	Result
Perturbation samples	2,500
Weighted local R squared	0.678
Random forest probability	0.691
Surrogate probability	0.666
Absolute local error	0.025

The surrogate agrees qualitatively with SHAP on the dominant features, but its weighted R squared of 0.678 and absolute probability error of 0.025 are not strong enough for quantitative reliance. The LIME-style result is therefore reported as corroborating evidence with a fidelity warning, not as a substitute for the random forest. This limitation illustrates why every local explanation requires its own quality check.

7 Synthetic Profile Dependence

Synthetic-profile dependence and conditional effects

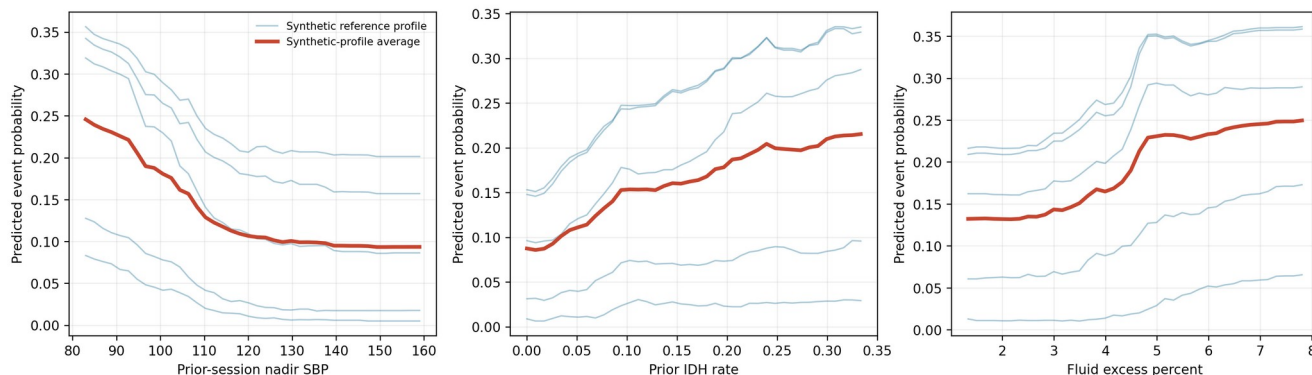


Figure 4. Dependence averages and conditional-effect curves across five synthetic reference profiles.

Average predicted risk falls sharply as prior-session nadir SBP rises from very low values. It increases with prior IDH rate and, more gradually, with fluid excess percentage. The spread of the synthetic-profile curves shows model heterogeneity without displaying any observed patient trajectory.

The curves vary one feature while leaving the other synthetic-profile values fixed. Because blood pressure, dry weight, fluid excess, and ultrafiltration measures are correlated, some combinations may be physiologically uncommon. These plots should be used to inspect model behavior, not to infer a safe treatment target or causal dose-response relationship.

8 Imbalance Leakage and Overfitting

Risk	Observed evidence	Assessment	Required response
Class imbalance	Test prevalence 8.495%; no-skill average precision equals prevalence	Material	Use average precision, sensitivity, precision, calibration, and workload metrics rather than accuracy alone
Patient leakage	Zero patient overlap across training, validation, and test partitions	Controlled in internal evaluation	Retain patient-disjoint splits and grouped cross-validation
Temporal leakage	Only index-time or lagged prior-session predictors are used; subsequent blood pressures define the outcome	Controlled by feature design	Preserve the prediction-time boundary in production feature pipelines
Overfitting or optimism	Grouped-CV AP 0.440, validation AP 0.470, test AP 0.395	Residual concern	Report patient-bootstrap intervals and require external temporal and geographic validation
Probability miscalibration	Test Brier score 0.062; calibration weakens in sparsely populated high-risk bins	Monitor	Recalibrate only on new local validation data and monitor calibration drift
Repeated-session dominance	21,354 test sessions arise from 170 patients	Material	Use patient-cluster bootstrap intervals and patient-equal-weight sensitivity analyses

Class imbalance is handled analytically rather than erased. Average precision, sensitivity, precision, calibration, and alert workload remain primary. Accuracy is not used to justify the model because predicting no event would already appear accurate for an outcome with approximately 8.5% prevalence.

Patient leakage is controlled by disjoint patient partitions and patient-grouped cross-validation. Temporal leakage is controlled by using only index-time or earlier-session predictors. Residual risks include patient-date linkage ambiguity, informative monitoring frequency, and the possibility that operational data fields would be recorded differently after deployment.

Overfitting remains plausible because the test average precision of 0.395 is lower than the validation value of 0.470 and because only 170 independent patients are in the test partition. Patient-bootstrap intervals, a locked test, and simpler comparators reduce optimism but do not replace external temporal and geographic validation.

9 Fairness Metrics

Metric	Definition	Ethical interpretation
Demographic parity difference	Maximum minus minimum alert-selection rate	Who receives an alert regardless of outcome
Disparate impact ratio	Minimum divided by maximum selection rate	Relative alert exposure; 0.80 is shown only as a screening heuristic
Equal opportunity difference	Maximum minus minimum sensitivity	Who receives the safety benefit among sessions with events
False-positive-rate difference	Maximum minus minimum FPR	Who bears unnecessary alerts among sessions without events
Equalized odds difference	Larger of sensitivity and FPR gaps	Worst conditional error-rate disparity
Predictive parity difference	Maximum minus minimum precision	Whether an alert has similar meaning across groups
Brier-score difference	Maximum minus minimum probability error	Whether probability quality differs across groups

The 0.80 ratio is a descriptive screening heuristic borrowed from another policy context, not a legal, regulatory, statistical, or clinical fairness certificate for DIAL-ALERT. Direction, uncertainty, prevalence, error severity, and workflow consequences must be reviewed together.

Fairness metrics diagnose different kinds of disparity

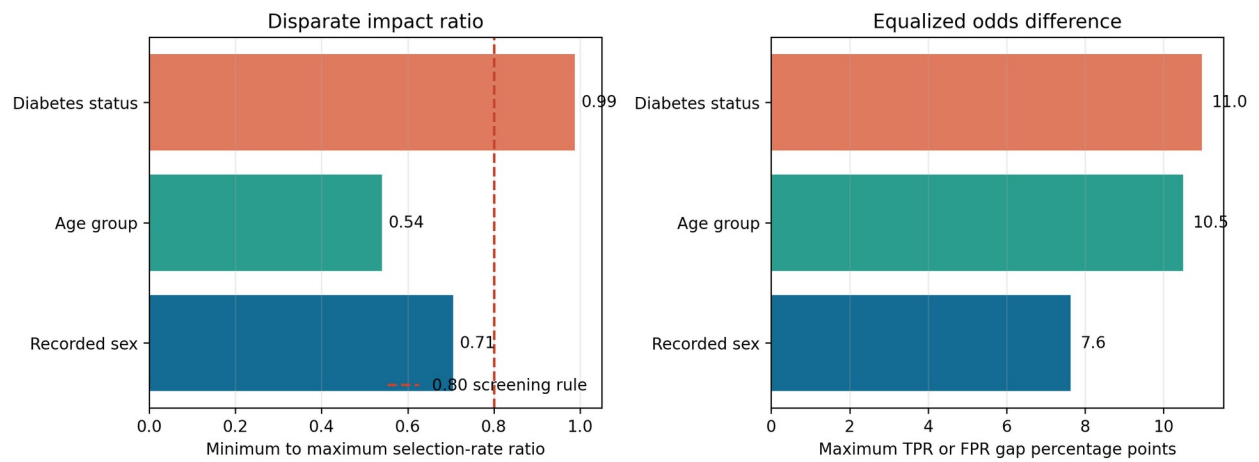


Figure 5. Disparate impact and equalized odds summarize different dimensions of subgroup disparity.

10 Subgroup Audit Results

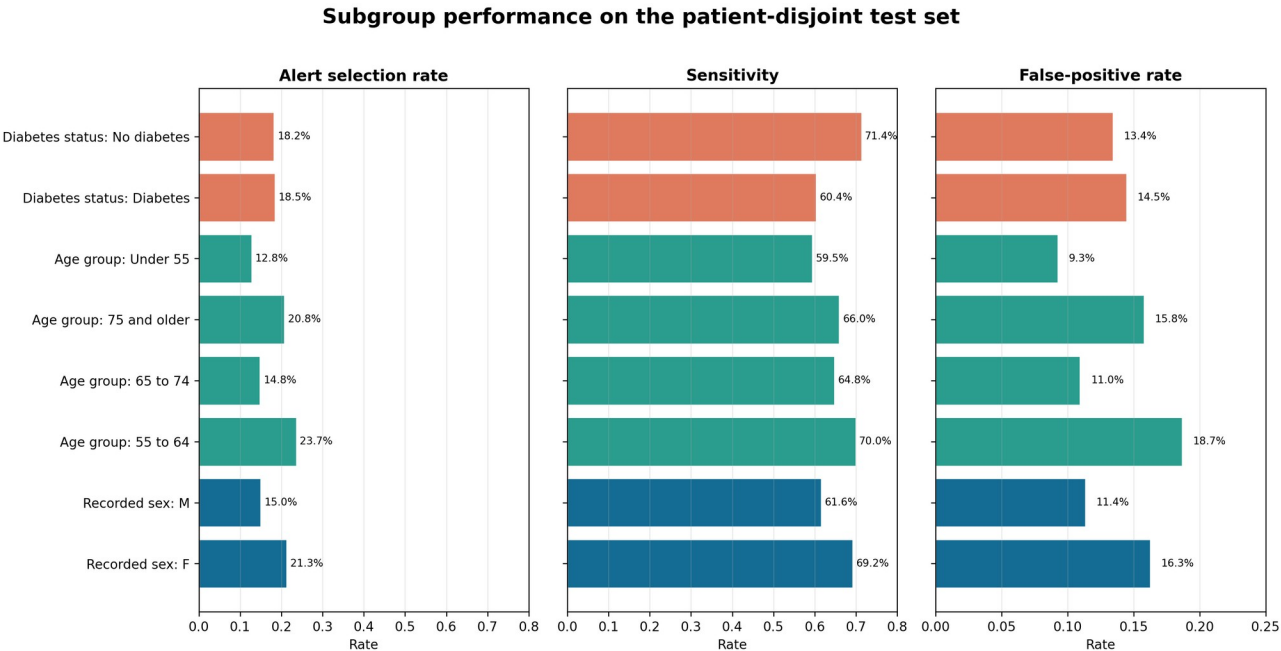


Figure 6. Alert selection sensitivity and false-positive rates by recorded sex age group and diabetes status.

Group	Patients	Prev	Alert	Sens	FPR	PPV
Recorded sex F	83	9.6%	21.3%	69.2%	16.3%	31.0%
Recorded sex M	87	7.3%	15.0%	61.6%	11.4%	30.0%
Age under 55	41	7.0%	12.8%	59.5%	9.3%	32.8%
Age 55 to 64	60	9.7%	23.7%	70.0%	18.7%	28.7%
Age 65 to 74	56	7.2%	14.8%	64.8%	11.0%	31.5%
Age 75 and older	39	9.9%	20.8%	66.0%	15.8%	31.5%
Diabetes	65	8.7%	18.5%	60.4%	14.5%	28.5%
No diabetes	105	8.3%	18.2%	71.4%	13.4%	32.5%

11 Interpretation of Subgroup Findings

Recorded sex. Female sessions had higher outcome prevalence (9.6% versus 7.3%), higher alert selection (21.3% versus 15.0%), higher sensitivity (69.2% versus 61.6%), and higher false-positive rate (16.3% versus 11.4%). The disparate impact ratio was 0.705, with a patient-bootstrap interval of 0.349 to 0.988. The equalized odds gap was 0.076, with interval 0.015 to 0.313. These wide intervals preclude a precise fairness conclusion.

Age. Alert selection ranged from 12.8% to 23.7%, producing a disparate impact ratio of 0.541. Sensitivity ranged from 59.5% to 70.0%. Age is both a predictor and a clinically relevant protected characteristic, so an apparent disparity can reflect prevalence, model behavior, threshold choice, or all three. It cannot be labelled unfair or acceptable without a clinical harm analysis.

Diabetes. Alert selection was almost identical (18.5% versus 18.2%), giving a disparate impact ratio of 0.988. Nevertheless, sensitivity was lower with diabetes (60.4% versus 71.4%) and average precision was also lower (0.347 versus 0.442). Selection parity alone would therefore miss an important performance gap.

Patient weighting and intersections. Giving each patient equal total weight changed several point estimates: recorded-sex disparate impact became 0.585, and the age equalized odds gap became 0.367. In the sex-by-age analysis, subgroup patient counts ranged from 19 to 32, and sensitivity ranged from 44.9% to 82.6%. These swings show that frequent attenders and small intersections materially influence results; they justify continued monitoring rather than subgroup-specific deployment claims.

12 Direct Dependence on Recorded Sex

A limited counterfactual diagnostic swapped F and M while holding all other predictors fixed. The mean absolute probability change was 0.007, the median was 0.005, and 1.35% of session classifications changed at the locked threshold. The largest individual probability change was 0.055.

This result suggests that the recorded-sex field has modest direct influence on most predictions. It does not show that the model is free from sex-related bias. Other predictors can encode correlated differences, and changing recorded sex while holding physiology fixed is not a biologically or socially valid causal intervention. The diagnostic must be read together with the observed subgroup error rates.

Diagnostic	Result
Mean absolute probability change	0.0070
Median absolute probability change	0.0048
Maximum absolute probability change	0.0549
Alert classifications changed	1.35%

13 Mitigation Experiments

Mitigation changes fairness and performance together

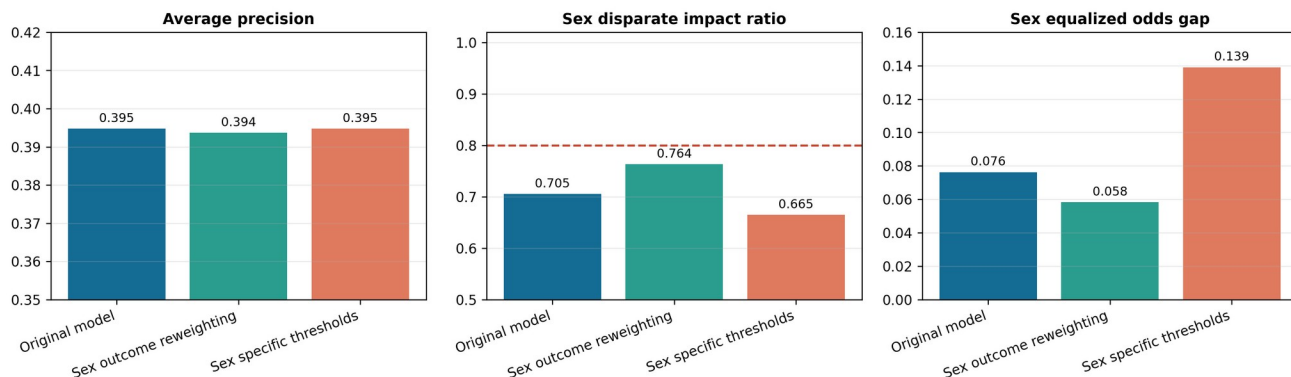


Figure 7. Overall performance and recorded-sex fairness metrics for two experimental mitigations.

Strategy	AP	Sens	PPV	DI ratio	EO gap
Original model	0.395	66.1%	30.6%	0.705	0.076
Sex-outcome reweighting	0.394	66.6%	30.9%	0.764	0.058
Sex-specific thresholds	0.395	68.5%	27.8%	0.665	0.139

Reweighting. Training rows were weighted so recorded-sex and outcome combinations contributed according to their marginal frequencies. This improved both recorded-sex fairness summaries with negligible overall performance change, but the disparate impact ratio remained below 0.80. Reweighting is the most promising tested option and should be re-evaluated with external data and broader attributes.

Sex-specific thresholds. Validation-tuned thresholds were 0.181 for F and 0.088 for M, targeting equal sensitivity on validation patients. On the test set, the equalized odds gap worsened from 0.076 to 0.139, precision fell from 30.6% to 27.8%, and the disparate impact ratio fell to 0.665. The approach failed to generalize and is rejected. It also requires a protected attribute at inference, which raises additional legal and ethical governance questions.

14 Mitigation and Governance Plan

Stage	Action	Purpose
Before modelling	Collect race ethnicity gender identity socioeconomic indicators symptoms treatments and site variables with consent and governance	Enables missing audits and improves target validity
Pre-processing	Evaluate patient-balanced and sex-outcome reweighting using validation data only	Reduces overrepresentation and selected error gaps
In-processing	Constrain sensitivity and FPR gaps during model selection when sample size supports it	Makes fairness part of optimisation rather than a late check
Post-processing	Use a single capacity-aware threshold by default; test group thresholds only under explicit governance	Avoids unstable or opaque differential treatment
Human factors	Display probability uncertainty key drivers data quality and a documented override path	Reduces automation bias and supports accountability
Monitoring	Track overall and subgroup calibration sensitivity precision FPR selection rate and alert burden	Detects drift and emergent disparity
Rollback	Suspend alerts when data drift calibration failure or subgroup harm thresholds are breached	Limits harm while the model is reviewed

A mitigation should not be accepted merely because one metric improves. The acceptance decision should require stable external results, preserved clinical utility, acceptable alert burden, transparent governance, and no material worsening in another subgroup or error type. DIAL-ALERT should remain decision support with accountable clinician override.

15 Ethical Risk Register

Hazard	Affected party	Severity	Control	Residual risk
Missed high-risk session	Patient	High	Clinician review, conservative sensitivity target, outcome monitoring	Residual false negatives remain
Alert fatigue	Nurses and physicians	High	Capacity-based alerting, configurable thresholds, burden dashboard	20% capacity still produces substantial false-alert burden
Unequal access to review	Protected or underserved groups	High	Subgroup monitoring, reweighting study, governance review	Race and socioeconomic fairness are unmeasured
Automation bias	Patients and clinicians	High	Decision-support-only label, explanation, override and documentation	Human factors require prospective testing
Privacy and re-identification	Patients	High	Minimum necessary data, access controls, audit logs, retention limits	Public data do not remove future deployment privacy duties
Performance drift	Future patients	Medium	Calibration and subgroup drift monitoring with rollback	No prospective drift evidence yet

Deployment Gates

- External temporal and geographic validation with patient-level uncertainty.
- Prospective silent-mode evaluation before any alert is shown to clinicians.
- Predefined subgroup sample-size and performance thresholds, including currently unavailable groups.
- Human-factors testing for alert comprehension, automation bias, alarm fatigue, and override documentation.
- Privacy, security, access-control, audit-log, incident-response, rollback, and model-change procedures.
- Clinical governance approval that specifies intended use, excluded uses, accountable owner, monitoring frequency, and stopping rules.

Until these gates are met, DIAL-ALERT should be presented only as a retrospective academic prototype. It must not determine treatment, deny care, or substitute for clinical assessment.

16 Conclusions

The audit identifies a clinically coherent but imperfect model. Aggregate global evidence and synthetic local explanations consistently emphasize prior haemodynamic instability and fluid status. Synthetic-profile dependence curves show nonlinear and heterogeneous behavior. The LIME-style surrogate supports the broad feature ranking but has only moderate local fidelity, so it should not be treated as a precise explanation.

Fairness conclusions depend on the metric. Recorded sex and age show notable alert-selection differences; diabetes has near-parity in selection but lower sensitivity and discrimination. Patient-bootstrap intervals are wide, patient-equal weighting changes point estimates, and small intersections are unstable. Race, ethnicity, socioeconomic status, and gender identity remain unaudited because the data do not contain them.

Sex-outcome reweighting is the only tested mitigation that improved both recorded-sex disparity summaries while preserving overall performance, although it did not eliminate disparity. Sex-specific thresholds failed on the held-out test patients and should not be used. The defensible decision is to continue research with broader data, external validation, patient and staff participation, and explicit deployment controls rather than claim fairness or clinical readiness.

Reproducibility

```
python src/audit_bias_fairness.py
python src/create_bias_fairness_report.py
```

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